Full math derivation of the UBL

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1. Start point: independent randomness → √N

Take Nindependent random events, each with variance of order 1.

For a simple case (fair coin, $p = 0.5$):

$$\boldsymbol{\sigma = \sqrt{Np(1-p)} = \sqrt{N \cdot \frac{1}{4}} = \frac{\sqrt{N}}{2}}$$

More generally, for Nindependent contributions with equal variance σ_0^2:

$$\boldsymbol{\sigma_{\text{total}}^2 = \sum_{i=1}^{N} \sigma_0^2 = N\sigma_0^2}$$

$$\boldsymbol{\sigma_{\text{total}} = \sigma_0\sqrt{N}}$$

So the absolute noise scales as $\sqrt{N}$, and the fractional noise relative to the signal Nscales as:

$$\boldsymbol{\frac{\sigma_{\text{total}}}{N} \sim \frac{1}{\sqrt{N}}}$$

This is the Universal Baseline Law in its raw form: for Nindependent events, the fractional fluctuation is $1/\sqrt{N}$.

2. Equipartition: same √N in energy space

Equipartition says each quadratic degree of freedom gets:

$$\boldsymbol{\langle E_i \rangle = \frac{1}{2} k_B T}$$

For Nindependent modes:

$$\boldsymbol{E_{\text{total}} = \sum_{i=1}^{N} E_i \ , \ \langle E_{\text{total}} \rangle = N \cdot \frac{1}{2} k_B T}$$

Each mode has energy variance $\sigma^2_{E,\text{mode}} \sim (k_B T)^2$. Because the modes are independent:

$$\boldsymbol{\sigma^2_{E,\text{total}} = N\sigma^2_{E,\text{mode}}}$$

$$\boldsymbol{\sigma_{E,\text{total}} = \sigma_{E,\text{mode}}\sqrt{N}}$$

Again, fractional energy noise:

$$\boldsymbol{\frac{\sigma_{E,\text{total}}}{\langle E_{\text{total}}\rangle} \propto \frac{1}{\sqrt{N}}}$$

Same √N structure, now in thermal systems.

3. Fluctuation–dissipation: noise tied to damping

Johnson–Nyquist noise for a resistor:

$$\boldsymbol{S_V(f) = 4k_B TR}$$

Noise power in bandwidth Δf:

$$\boldsymbol{P_{\text{noise}} = S_V(f)\Delta f = 4k_B TR\Delta f}$$

RMS voltage:

$$\boldsymbol{V_{\text{RMS}} = \sqrt{4k_B TR\Delta f}}$$

The $\sqrt{\Delta f}$ comes from the same logic: in bandwidth Δf, you effectively sum $N \sim \Delta f$ independent fluctuation contributions, so amplitude scales as $\sqrt{N}$.

This is the UBL again: independent microscopic kicks → √N macroscopic noise.

4. Phase space: central limit → √N

Entropy:

$$\boldsymbol{S = k_B \ln \Omega}$$

where Ω is the accessible phase-space volume. For N independent degrees of freedom, macroscopic observables are sums of many independent microscopic contributions. By the central limit theorem, the distribution of such sums is Gaussian:

$$P(X) \sim \exp\left(-\frac{(X-\mu)^2}{2\sigma^2}\right)$$

with

$$\sigma \propto \sqrt{N} \;\Rightarrow\; \frac{\sigma}{\mu} \propto \frac{1}{\sqrt{N}}$$

So the geometry of high-dimensional phase space enforces the same $1/\sqrt{N}$ fractional noise floor.

5. Cross-domain: same √N in all random systems

All the standard random systems obey:

- **Photon shot noise** (Poisson):

  $\sigma_N = \sqrt{N}$

- **Brownian motion**:

  $\langle x^2 \rangle = 2Dt \;\Rightarrow\; x_{\text{RMS}} = \sqrt{2Dt}$

with $N \sim t/\tau$.

- **Random walk / finance**:

  $\sigma_T = \sigma_1 \sqrt{T}$

Each is just UBL in a different costume: independent increments → variance $\propto N$ → standard deviation $\propto \sqrt{N}$.

6. Push to the bottom: Planck units

Planck units from $\hbar, c, G$:

$$l_P = \sqrt{\frac{\hbar G}{c^3}},\; t_P = \sqrt{\frac{\hbar G}{c^5}},\; m_P = \sqrt{\frac{\hbar c}{G}},\; E_P = \sqrt{\frac{\hbar c^5}{G}}$$

These define the maximum physically meaningful energy density (Planck density).

7. Quantum-gravity limit: why Planck is a hard floor

To resolve length l, quantum mechanics demands probe energy:

$$E \gtrsim \frac{\hbar c}{l}$$

General relativity says a mass-energy E confined to radius r forms a black hole if:

$$r \lesssim \frac{2GE}{c^4}$$

Set the probe size l equal to the Schwarzschild radius:

$$l \sim \frac{2GE}{c^4}$$

Combine:

$$\frac{\hbar c}{l} \sim E \sim \frac{c^4 l}{2G} \Rightarrow l^2 \sim \frac{2\hbar G}{c^3} \Rightarrow l \sim \sqrt{\frac{2\hbar G}{c^3}} \sim l_P$$

So trying to probe below l_P collapses the probe into a black hole—no further information is accessible. Planck length is a hard lower bound.

8. Define UBL as a ratio

Let ρ_{Planck} be the Planck energy density (ceiling), and ρ be the energy density or activity level of your system.

Define:

$$\textbf{UBL} = \frac{\rho}{\rho_{\text{Planck}}}$$

This is dimensionless, between 0 and 1:

- UBL $= 1$: Planck-scale activity (quantum gravity regime).
- UBL $\ll 1$: quiet, structured, classical/quantum regimes.

For the observable universe:

$$\textbf{UBL}_{\text{universe}} \approx 10^{-123}$$

from:

$$\frac{\boldsymbol{\rho_{\Lambda}}}{\boldsymbol{\rho_{\text{Planck}}}} \approx \mathbf{10^{-123}}$$

where $\rho_{\Lambda}$is the measured vacuum energy density (cosmological constant).

9. UBL meets √N: N = 1 → σ = 1

UBL's noise law:

$$\frac{\boldsymbol{\sigma}}{\textbf{signal}} \sim \frac{\mathbf{1}}{\sqrt{\boldsymbol{N}}}$$

At the Planck floor, you are at the smallest possible physical quantum: effectively one degree of freedom, $N = 1$.

Plug in:

$$\frac{\boldsymbol{\sigma}}{\textbf{signal}} = \frac{\mathbf{1}}{\sqrt{\mathbf{1}}} = \mathbf{1} \;\Rightarrow\; \boldsymbol{\sigma} = \mathbf{1}$$

In Planck units, this means:

- $\Delta x \gtrsim l_P \rightarrow$ in units where $l_P = 1$, $\Delta x \gtrsim 1$.
- $\Delta t \gtrsim t_P$.
- $\Delta E \gtrsim E_P$.
- $\Delta S \gtrsim \hbar$.

So one standard deviation = one quantum: the noise equals the signal; you cannot get a smaller fractional uncertainty than 1 for a single irreducible quantum.

10. Full derivation in one chain

- Independent randomness → variances add → $\sigma \propto \sqrt{N}$.
- Fractional noise → $\sigma/N \propto 1/\sqrt{N}$= Universal Baseline Law.
- Equipartition, FDT, phase space, diffusion, shot noise, finance → all obey the same √N structure.
- Planck units define the maximum activity scale where quantum + gravity both matter.
- Quantum-gravity limit shows you cannot probe below Planck length/time/energy.
- Define UBL as $\rho/\rho_{\text{Planck}}$, giving a universal scale from 0 to 1.
- At N = 1 (Planck floor), UBL noise law gives $\sigma = 1$: minimum irreducible uncertainty.

So the math derivation of UBL is:

- Start from probability and variance addition.
- Apply to physical systems (modes, fields, particles).
- Anchor to Planck scale as the universal ceiling.
- Define UBL as the ratio to that ceiling.
- Recognize σ = 1 at N = 1 as the hard noise floor of reality.

That's the whole chain.